



STUDENT NAME: _____

CHEN / QUINNELL

Prairiewood High School

2024 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 1

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Write your name on each writing booklet

Total marks: 70

Section I – 10 marks (pages 3 - 7)

- Attempt Questions 1 - 10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 8 - 14)

- Attempt Questions 11 - 14
- Allow about 1 hour and 45 minutes for this section

MC	Q11	Q12	Q13	Q14	Total
/ 10	/ 15	/ 15	/ 15	/ 15	/ 70

DO NOT REMOVE ANY SECTION OF THIS EXAM FROM THE EXAMINATION ROOM

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SECTION I: Multiple Choice (10 marks)

Attempt Questions 1 – 10.

Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for this section.

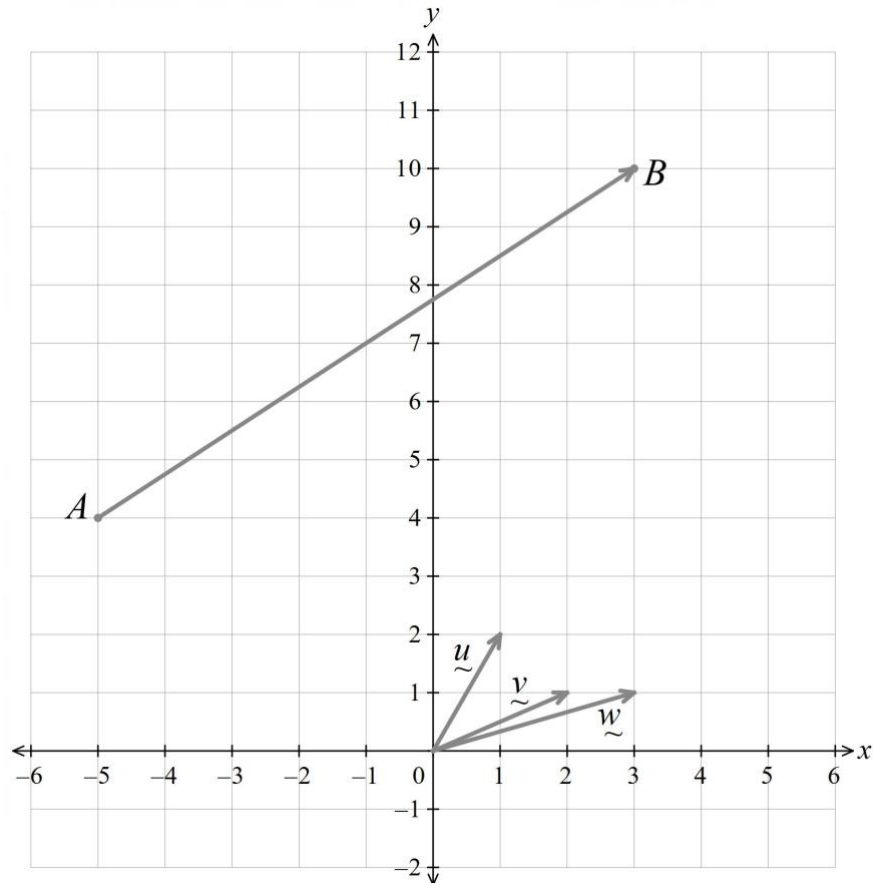
1. What is the remainder when the polynomial $P(x) = x^3 - 3ax$ is divided by $(x - a + 1)$?
- (A) $a^3 + 1$
(B) $a^3 + 6a - 1$
(C) $a^3 - 6a^2 - 1$
(D) $a^3 - 6a^2 + 6a - 1$
2. A bag contains 6 blue balls, 7 green balls, 8 yellow balls, 9 orange balls and 10 red balls. Balls are drawn at random one at a time without replacement from the bag. What is the least number of balls that needs to be drawn from the bag in order to be certain that 8 balls of the same colour are amongst those selected?
- (A) 21
(B) 22
(C) 35
(D) 36
3. What is the Cartesian equation of the curve with the parametric equations below?
- $$x = 3\sin\theta + 1$$
- $$y = 3\cos\theta$$
- (A) $x^2 + (y - 1)^2 = 3$
(B) $(x - 1)^2 + y^2 = 3$
(C) $x^2 + (y - 1)^2 = 9$
(D) $(x - 1)^2 + y^2 = 9$

4. Let X be a random variable that follows a Binomial distribution with $E(X) = 7$ and $Var(X) = 6$. Which of the following could be the probability of success p ?

- (A) $\frac{1}{49}$
 (B) $\frac{1}{7}$
 (C) $\frac{36}{49}$
 (D) $\frac{6}{7}$

5. On the diagram below, the vector $\overrightarrow{AB}$ is shown joining $A(-5,4)$ and $B(3,10)$. There are three other vectors shown on the diagram:

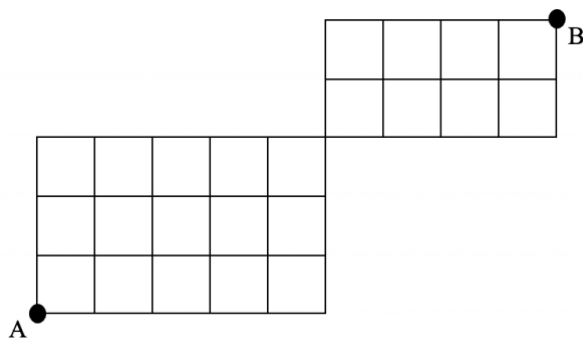
$$\underline{u} = \underline{i} + 2\underline{j} \quad \underline{v} = 2\underline{i} + \underline{j} \quad \underline{w} = 3\underline{i} + \underline{j}$$



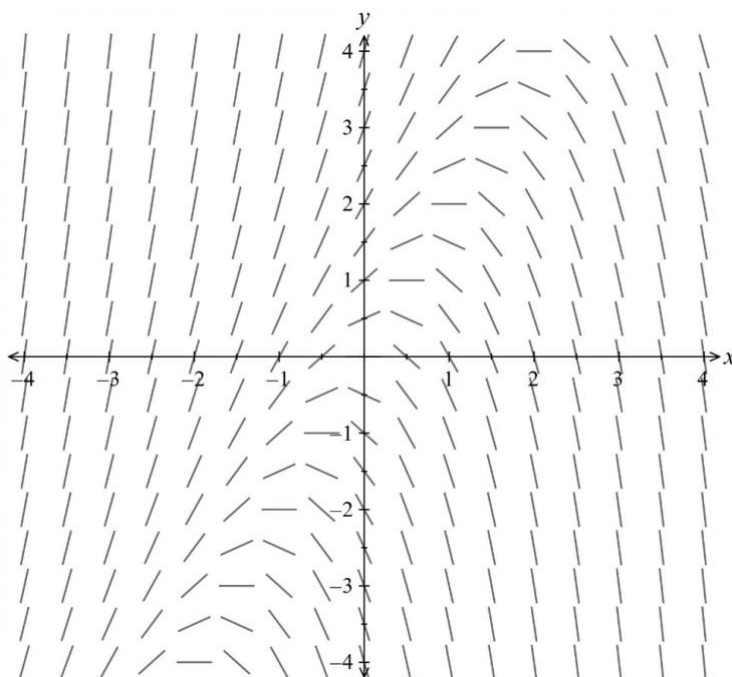
Which of the following vectors is parallel to $\overrightarrow{AB}$?

- (A) $\underline{u} - \underline{v}$
 (B) $\underline{u} + \underline{v}$
 (C) $\underline{w} + \underline{v}$
 (D) $\underline{w} + \underline{u}$

6. Moving only right or up, how many different paths exist to get from Point A to Point B ?



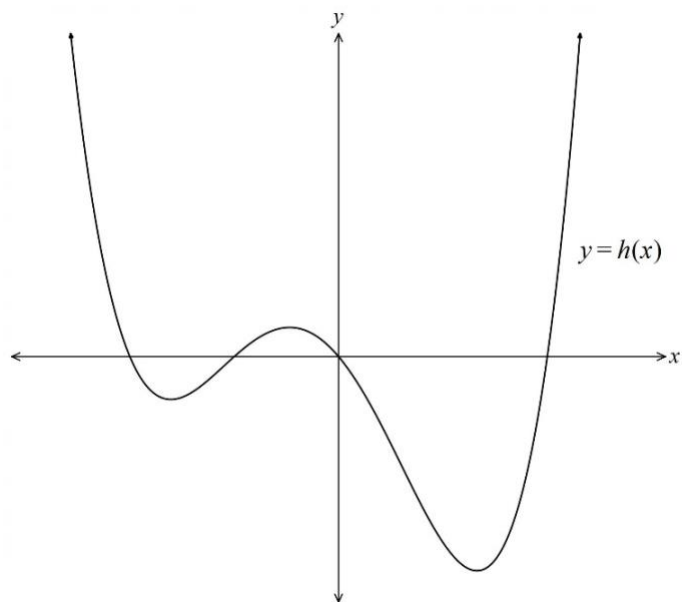
- (A) 45
 (B) 840
 (C) 480480
 (D) 16!
7. The direction field for a differential equation is shown below.



Considering the point (1, 1) determine which of the differential equations below could give this direction field?

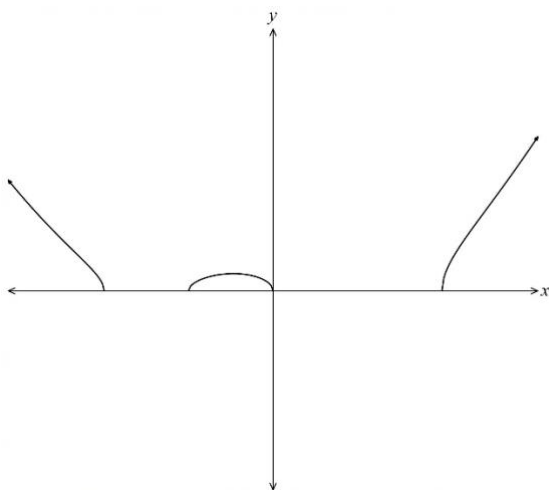
- (A) $y' = (y - 2x)$
 (B) $y' = (y + 2x)$
 (C) $y' = (y + x)$
 (D) $y' = (x - y)$

8. The graph of $y = h(x)$ is shown below.

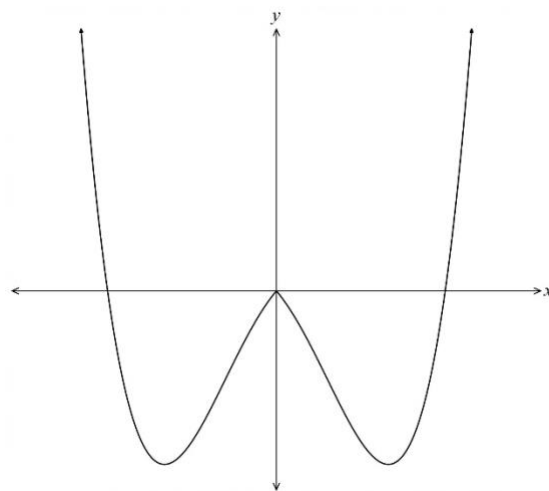


Which graph shows $y = |h(x)|$?

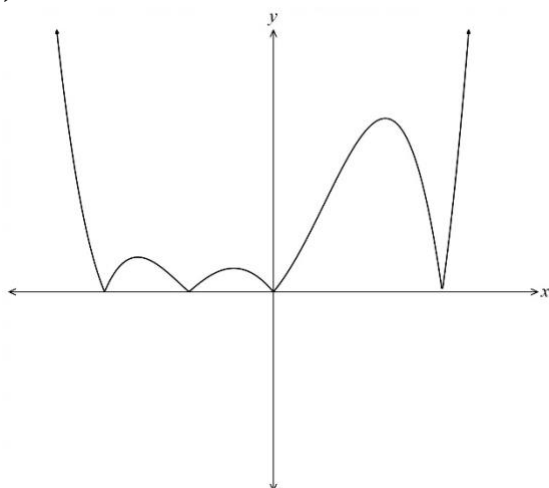
(A)



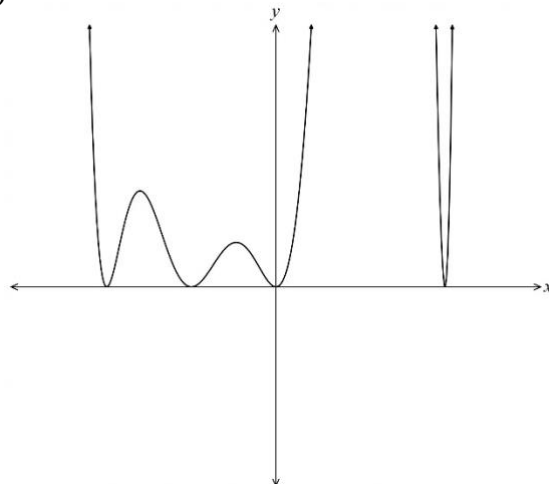
(B)



(C)



(D)



9. If $y = \tan^{-1}\left(\frac{1}{x}\right)$, which of the following is an expression for $\frac{dy}{dx}$?

(A) $\frac{dy}{dx} = -\sin^2 y$

(B) $\frac{dy}{dx} = -\cos^2 y$

(C) $\frac{dy}{dx} = -\tan^2 y$

(D) $\frac{dy}{dx} = -\cot^2 y$

10. What is the range of the function $y = 2 \sin x + \cos x - 1$?

(A) $[-1 - \sqrt{5}, -1 + \sqrt{5}]$

(B) $[1 - \sqrt{5}, 1 + \sqrt{5}]$

(C) $[-\sqrt{5}, \sqrt{5}]$

(D) $[-3, 1]$

End of Section I

SECTION II (60 marks)**Attempt Questions 11 – 14.****Allow about 1 hours and 45 minutes for this section.****Answer each question in the appropriate writing booklet. Extra booklets are available.****For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.**

Question 11 (15 marks) Start a new writing booklet.

- (a) Find the number of ways that the letters of the word TRIANGLE can be arranged in a line such that the vowels are all next to each other but the consonants are not all next to each other. 2

- (b) Prove that $\frac{\sin \theta}{\cos \theta + \sin \theta} + \frac{\sin \theta}{\cos \theta - \sin \theta} = \tan(2\theta)$. 2

- (c) Find the exact value of the integral below: 2

$$\int_0^1 \frac{2x^2 + 1}{25 + (2x^3 + 3x)^2} dx$$

- (d) Use the $t = \tan \frac{x}{2}$ results to show that $\sin x - \tan \frac{x}{2} = \tan \frac{x}{2} \cos x$ 2

- (e) Given α, β and γ are the roots of the equation $4x^3 + 5x^2 + 4x - 1 = 0$, find the value of: 2

$$\frac{1}{\alpha^2 \beta^2} + \frac{1}{\beta^2 \gamma^2} + \frac{1}{\alpha^2 \gamma^2}$$

Question 11 continues on the next page.

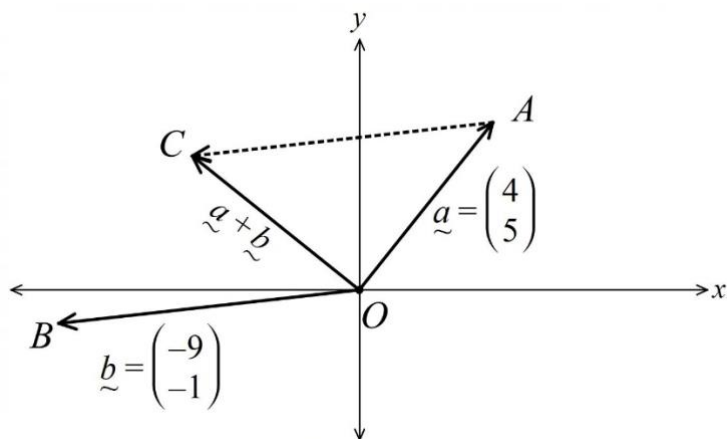
Question 11 (continued)

(f) Solve $\frac{1}{2x-x^2} \geq 1$.

3

(g) The diagram shows the vectors $\vec{a} = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$, $\vec{b} = \begin{pmatrix} -9 \\ -1 \end{pmatrix}$ and the resultant vector $\vec{a} + \vec{b}$.

2



Show that $\triangle OAC$, formed by the vectors, is an isosceles triangle and find the size of $\angle OAC$.

End of Question 11

Question 12 (15 marks) Start a new writing booklet.

- (a) On the same diagram, sketch the curves $y = \frac{1}{\sin^{-1} x}$ and $y = \frac{1}{\cos^{-1} x}$, 3

clearly showing any intercepts, endpoints and the equations of any asymptotes.

- (b) Given that $P(x) = x^4 - 9x^3 + 25x^2 - 27x + 10$, has a double root, factorise $P(x)$ into linear factors. 2

- (c) A thin sheet of ice is in the form of a circle. If the ice is melting in such a way that the area of the sheet is decreasing at a rate of $0.5 \text{ m}^2/\text{s}$. At what rate is the radius decreasing when the area of the sheet is 12 m^2 ? 3

- (d) Express $\sqrt{6} \sin x + \sqrt{2} \cos x$ in the form $R \sin(x + \alpha)$ and hence or otherwise, solve: 3

$$\sqrt{6} \sin x + \sqrt{2} \cos x = 2, \quad [0, 2\pi]$$

- (e) Show that $\sin^{-1} x + \sin^{-1} \sqrt{1 - x^2} = \frac{\pi}{2}$. 2

- (f) Using the substitution $u = -\frac{1}{x}$, evaluate 2

$$\int_1^2 \frac{1}{x^2} e^{-\frac{1}{x}} dx.$$

End of Question 12

Question 13 (15 marks) Start a new writing booklet.

- (a) A box initially contains 100 grams of bacteria and 10 000 grams of a radioactive material with a half-life of 3 hours. An hour later, the mass of bacteria has grown to 150 grams. Assuming exponential growth and decay, what time (t) will the mass of bacteria and radioactive material be the same? Leave your answer correct to 2 decimal places. 3
- (b) Use the method of separation of variables to solve the differential equation: $\frac{dy}{dx} = \frac{3y}{x}$, given that $\frac{dy}{dx} = 6$ when $x = 2$. 2
- (c) Records show that 40% of residents of a town went to school within the town. A sample of 150 residents of the town is to be taken to determine the population who went to school within the town.
- (i) Find, in simplest fraction form, the mean and standard deviation of the distribution of such sample proportions. 2
- (ii) Use the following extract of the table of values of $P(Z \leq z)$, where Z has a standard normal distribution, to estimate correct to 2 decimal places the probability that a sample of 150 residents of the town contains at least 57 and at most 69 who went to school in town. 3

STANDARD NORMAL DISTRIBUTION TABLE

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767

Question 13 continues on the next page.

Question 13 (continued)

- (d) Use mathematical induction to prove that $9^n - 4^n$ is divisible by 5 for all integers $n \geq 1$. **3**
- (e) Given that ${}^nC_4 = p$ and ${}^nC_5 = q$, use the Pascals Triangle relationships between coefficients to find an expression, in terms of p and q , for ${}^{n+1}C_{n-5} + {}^nC_{n-4}$. **2**

End of Question 13

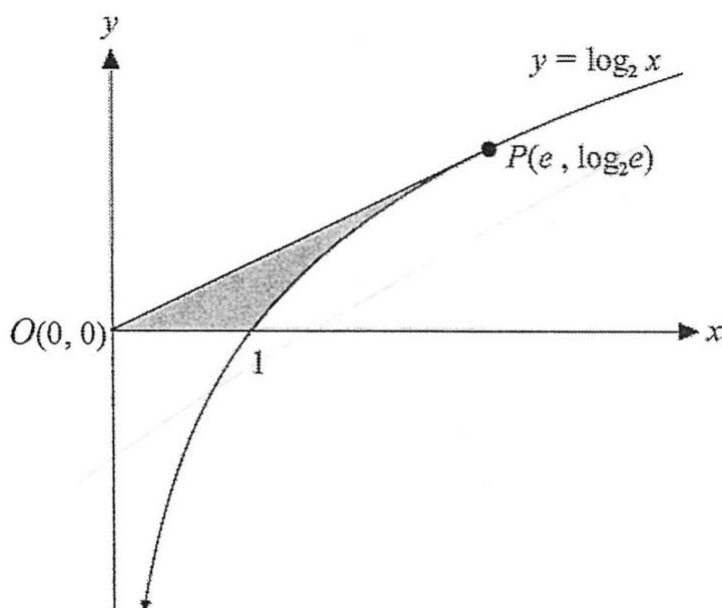
Question 14 (15 marks) Start a new writing booklet.

- (a) In the expansion of the expression $(1 + x)(a - bx)^{12}$, the coefficient of x^8 is zero. 2
Find $\frac{a}{b}$ in simplest form provided neither a or b are zero.

- (b) Use Mathematical Induction to show that for all integers $n \geq 1$ 3

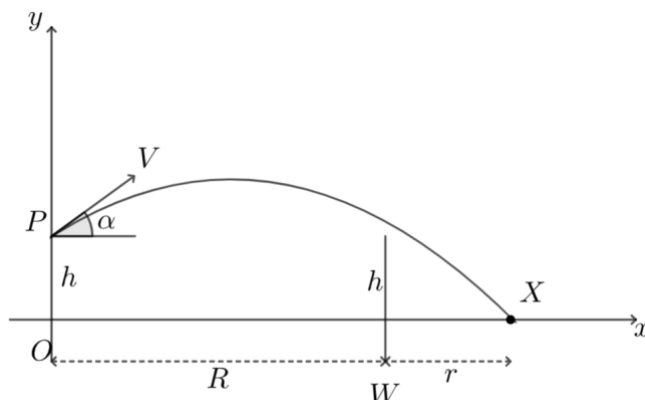
$$\left(1 \times 1! \times \frac{0}{1!}\right) + \left(2 \times 2! \times \frac{1}{2!}\right) + \left(3 \times 3! \times \frac{2}{3!}\right) + \cdots + \left(n \times n! \times \frac{n-1}{n!}\right) = (n+1)! - \frac{1}{n!}$$

- (c) The diagram shows the point $P(e, \log_2 e)$ on the curve $y = \log_2 x$. 3
 OP is a tangent to the curve at the point P .



The shaded region bounded by the curve $y = \log_2 x$, the line OP and the x axis is rotated through one revolution about the y axis. Find, in simplest exact form, the volume of the solid formed.

- (d) A projectile is launched from point P at a height of h metres above the ground with velocity V m/s at an angle α to the horizontal towards the point X . There is a wall with height h metres at point W , which is R metres from the point O .



The position vector of the projectile, t seconds after it is projected and with acceleration due to gravity, g , is given by

$$\mathbf{r}(t) = \begin{pmatrix} Vt \cos \alpha \\ -\frac{1}{2}gt^2 + Vt \sin \alpha + h \end{pmatrix}$$

DO NOT prove this.

- (i) The projectile just clears the wall. Show that

3

$$V^2 \geq \frac{gR}{2 \sin \alpha \cos \alpha}.$$

- (ii) After clearing the wall, the projectile hits the ground at the point X . Show that

2

$$\tan \alpha \geq \frac{Rh}{(R+r)r}.$$

- (iii) Suppose the projectile clears the wall, and $V \leq 50$, $g = 10$, $R = 80$, and $h = 1$. What is the closest point to the wall where the projectile can land?

2

END OF EXAMINATION

Student Name: _____

Teacher: CHEN / QUINNELL

Section I: Multiple Choice Answer Sheet

Completely fill in the most correct answer

1 A ☐ B ☐ C ☐ D ☐

2 A ☐ B ☐ C ☐ D ☐

3 A ☐ B ☐ C ☐ D ☐

4 A ☐ B ☐ C ☐ D ☐

5 A ☐ B ☐ C ☐ D ☐

6 A ☐ B ☐ C ☐ D ☐

7 A ☐ B ☐ C ☐ D ☐

8 A ☐ B ☐ C ☐ D ☐

9 A ☐ B ☐ C ☐ D ☐

10 A ☐ B ☐ C ☐ D ☐